| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2018 | **Task Last Updated:** | FY 08/06/2018 |
| **PI Name:** | Kirsch, David M.D., Ph.D. | | |
| **Project Title:** | Defining the Role of miR-182 in HZE-Induced Tumorigenesis | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | david.kirsch@duke.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 919-681-8586 |
| **Organization Name:** | Duke University | | |
| **PI Address 1:** | 450 Research Dr | | |
| **PI Address 2:** | Box 91006, LSRC Bldg. Room B230 | | |
| **PI Web Page:** | | | |
| **City:** | Durham | **State:** | NC |
| **Zip Code:** | 27710-0001 | **Congressional District:** | 1 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2013 Space Radiobiology NNJ13ZSA001N |
| **Start Date:** | 01/06/2014 | **End Date:** | 01/05/2018 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Simonsen, Lisa | **Contact Phone:** | |
| **Contact Email:** | lisa.c.simonsen@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNX14AC96G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | Important opportunities for space travel in the decades ahead include a proposed mission to Mars. However, an important deterrent to such interplanetary travel is the risk of radiation-induced carcinogenesis. There are many uncertainties in predicting the risk of space travel because the mechanisms of carcinogenesis from space radiation, which is dominated by high-energy protons and high (H) charge (Z) and energy (E) (HZE) particles, are poorly understood. Here, we will test the hypothesis that microRNAs that regulate the expression of proteins that are known to regulate radiation-induced cancer will function as non-protein encoding genetic factors that mediate radiation carcinogenesis from HZE particles. To test this hypothesis, we will utilize novel genetically engineered mice in which we can delete or overexpress miR-182. Mice will be exposed to low LET (linear energy transfer) and HZE particle radiation. Based on our preliminary experiments, we anticipate that mice lacking miR-182 will be protected from radiation-induced cancer, while mice overexpressing miR-182 will be sensitized to radiation carcinogenesis. Therefore, | | |

| | |
|---|---|
| | we expect that this research will establish that miRNAs are nonprotein encoding genetic factors that regulate radiation carcinogenesis. Furthermore, by studying how radiation quality impacts radiation-induced cancer in the setting of miR-182 deletion and overexpression, we will gain new insights into the mechanisms of the relative biological effectiveness (RBE)/LET relationship for carcinogenesis. Ultimately, this information can be used to define better risk estimates of cancer from deep space missions. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Cancer is a major cause of morbidity and mortality. Cancer arises from a combination of genetic mutations in tumor cells and alterations in the microenvironment. Further research into the mechanism of cancer development is needed so that more successful strategies for prevention and treatment of cancer can be developed. By studying the mechanisms of cancer initiation and development, we will generate new knowledge that can be used to develop novel approaches for the prevention and treatment of cancer. Here, we will study the role of miRNA 182 in radiation-induced cancer. Therefore, this work will help define the role of miRNAs in radiation-induced cancer. |
| **Task Progress:** | During the funding period, we made significant progress towards determining the role of microRNA-182 (miR-182) in radiation carcinogenesis. We bred, infected, and irradiated various strains of genetically engineered mice to assess the effect of miR-182 loss or overexpression on tumorigenesis. These mice were treated with multiple doses and fractionations of X-rays at Duke University and 56Fe and 28Si particles at Brookhaven National Laboratory (BNL). In addition, members of our lab interacted with outside experts in space radiation and radiation-induced tumorigenesis by presenting at the NASA Human Research Program Investigators' Workshop in Galveston, Texas each year during the funding period. Furthermore, two of the graduate students in the Kirsch laboratory successfully completed the NASA Space Radiation Summer School. Through these activities and from our research projects, we have gained new information about the effects of microRNAs on regulating tumorigenesis in response to terrestrial and space radiation. We tested the hypothesis that microRNAs that control the expression of proteins known to regulate radiation-induced cancer will function as genetic factors in radiation carcinogenesis. In order to study the role of microRNA-182 (miR-182) in radiation-induced lymphomagenesis and lung tumorigenesis following exposure to X-ray or space irradiation, we developed a mouse model that lacks miR-182 expression (Aim 1) or expresses an elevated level of miR-182 (Aim 2).<br><br>For Aim 1, we irradiated miR-182 knockout mice with a fractionated dose of X-rays in order to investigate the role of miR-182 in radiation-induced lymphomagenesis. However, we did not observe a significant impact of miR-182 expression level on the latency or the incidence of radiation-induced lymphomas.<br><br>We also crossed our mouse model of lung cancer (RbFL/FL; p53FL/+) to mice that lack miR-182 in the whole animal. In the absence or presence of radiation, we found that loss of miR-182 reduced overall survival when compared to littermates that retained wild type expression of miR-182. From these data, we conclude that miR-182 functions to regulate the survival of lung-tumor bearing mice. However, we have not seen evidence that there is a significant interaction between miR-182 expression and radiation exposure on survival. We are currently working with a veterinary pathologist to specifically characterize the impact of miR-182 knockout and radiation exposure on lung tumor burden and aggressiveness.<br><br>For Aim 2, we irradiated mice that overexpress miR-182 in the whole animal with a fractionated dose of X-rays to investigate the role of miR-182 in radiation-induced lymphomagenesis. Consistent with our hypothesis that miR-182 promotes radiation-induced tumorigenesis, mice overexpressing miR-182 developed lymphomas at an accelerated rate post radiation exposure. We are quantifying Fbxw7 and c-Myc expression and protein levels within these radiation-induced lymphomas to characterize the mechanism through which miR-182 overexpression promotes the development of lymphomas.<br><br>We also crossed our mouse model of lung cancer (RbFL/FL; p53FL/+) to mice that overexpress miR-182 in the whole animal. In contrast to the data in the miR-182 knockout mouse model, overexpression of miR-182 did not significantly impact overall survival of lung tumor-bearing mice in the presence or absence of radiation exposure. We are currently working with a veterinary pathologist to specifically characterize the impact of miR-182 expression and radiation exposure on lung tumor burden and aggressiveness. |
| **Bibliography Type:** | Description: (Last Updated: 11/19/2025) |
| **Abstracts for Journals and Proceedings** | Castle KD, Lee CL, Sachdeva M, Moding EJ, Williams N, Ma Y, Luo L, Kirsch DG. "Defining the role of miR-182 in radiation-induced lymphomagenesis and lung tumorigenesis." 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018.<br>2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018 |
| **Abstracts for Journals and Proceedings** | Castle KD, Lee CL, Sachdeva M, Moding EJ, Williams N, Ma Y, Luo L, Kirsch DG. "Defining the role of miR-182 in radiation-induced lymphomagenesis and lung tumorigenesis." 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017.<br>2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017 |
| **Abstracts for Journals and Proceedings** | Castle KD, Lee CL, Sachdeva M, Moding EJ, Williams N, Ma Y, Luo L, Kirsch DG. "Defining the role of miR-182 in radiation-induced lymphomagenesis and lung tumorigenesis." 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016.<br>2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016 |